

SET-B

Q1) $\begin{array}{ccccccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \end{array}$
 $\underbrace{\hspace{1.5cm}}_{\text{MSB}=1} \therefore \text{no. is -ve}$

invert $\rightarrow \begin{array}{ccccccc} 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{array}$
 $+ \begin{array}{ccccccc} & & & & & & & 1 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{array} = 8 \text{ in decimal}$

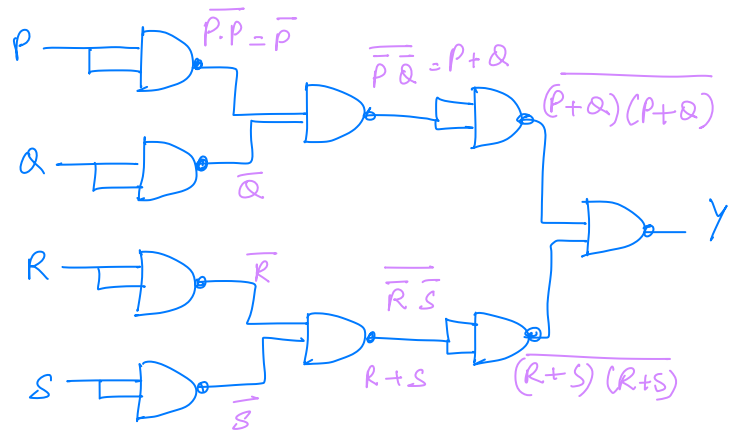
$\therefore -8$

Q1: -8

Q2)

$P + Q + R + S$

$$\begin{aligned} Y &= \overline{(P+Q)(P+Q) \cdot (R+S)(R+S)} \\ &= \overline{(P+Q) \cdot (R+S)} \\ &= P+Q + R+S \end{aligned}$$



Q3) $(101011)_2 = 43$

Q4) $7.96 \times 10^6 \text{ A/Wb}$

$$\begin{aligned} R &= \frac{l}{\mu_0 \mu_r A} \\ &= \frac{4}{(4\pi \times 10^{-7})(80)(0.005)} \\ &\approx 7.96 \times 10^6 \text{ A/Wb} \end{aligned}$$

Q5) 13.4 A

$$A = 500 \text{ mm}^2 = 500 \times 10^{-6} \text{ m}^2, \quad l = 400 \text{ mm} = 400 \times 10^{-3} \text{ m}$$

$$\phi = 800 \times 10^{-6} \text{ Wb}, \quad N = 100, \quad \mu_r = 380$$

$$B = \frac{\phi}{A}, \quad H = \frac{B}{\mu_0 \mu_r}, \quad F = NI = Hl$$

$$I = \frac{Hl}{N} = \frac{3353.9 \times 400 \times 10^{-3}}{100}$$

$$I \approx 13.4 \text{ A}$$

$$\frac{800 \times 10^{-6}}{500 \times 10^{-6}} = \frac{1.6 \text{ T}}{(4\pi \times 10^{-7})(380)} = 3353.9 \text{ A/m}$$

Q6) 10V

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$V_s = V_p \times \frac{N_p}{N_s} = 200 \text{ V} \times \frac{10}{200} = 10 \text{ V}$$

Q7) 32 A

$$S = V_1 I_1$$

$$I_1 = \frac{S}{V_1} = \frac{80 \times 10^3 \text{ VA}}{2500 \text{ V}} = 32 \text{ A}$$

$$V_1 = 2.5 R = 2500$$

Q8) → C

$$\frac{E(s)}{R(s)} = ?$$

$$G = \frac{C(s)}{E(s)}$$

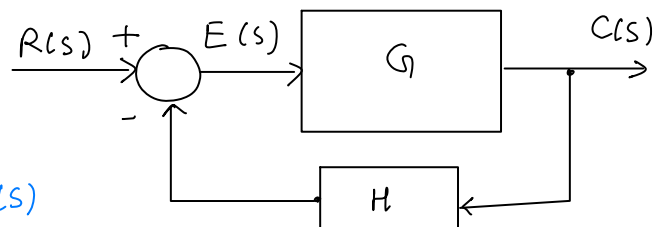
$$E(s) = R(s) - H \times C(s)$$

$$E(s) = R(s) - H \times G \times E(s)$$

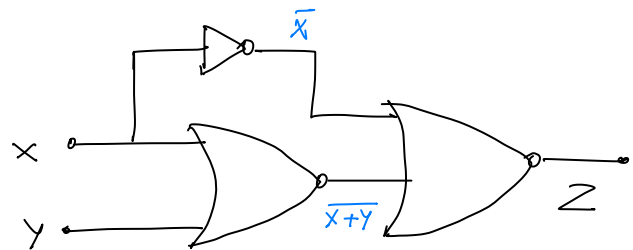
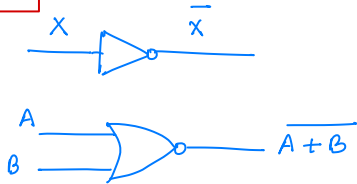
$$E(s) + H \times G \times E(s) = R(s)$$

$$E(s) \{ 1 + HG \} = R(s)$$

c) $\frac{E(s)}{R(s)} = \frac{1}{1 + GH}$



Q9) - A



$$\begin{aligned}
 Z &= \overline{\bar{X} + \overline{X+Y}} \\
 &= \overline{\bar{X} + \bar{X} \cdot \bar{Y}} \quad \text{De Morgan's law} \\
 &= (\overline{\bar{X}}) \cdot (\overline{\bar{X} \cdot \bar{Y}}) \\
 &= X \cdot (\overline{\bar{X} + \bar{Y}}) = X \cdot (X + Y) = X \quad \text{Absorption law}
 \end{aligned}$$

$$Z = \bar{X} + \overline{X+Y}$$

A) $Z = X$

Q10

$$XY + XZ' + YZ$$

Q10 - C

$$YZ + XZ' + XY(Z + Z') \quad \because Z + Z' = 1$$

$$YZ + X\bar{Z}$$

$$YZ + XYZ + XZ' + XYZ'$$

$$= YZ(1 + X) + XZ'(1 + Y)$$

$$\{1 + A = 1\}$$

$$= YZ + XZ'$$

C) $YZ + XZ'$

Q11) B)

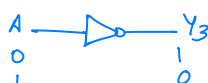
NOR or X-NOR



$$Y = \overline{A+B}$$



NOT X



AND



OR



NAND



XOR

A	B	Y1
0	0	1
0	1	0
1	0	0
1	1	0

Y2
1
0
0
1

Y4
0
0
0
1

Y5
0
1
1
1

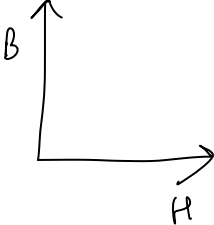
Y6
1
1
1
0

Y7
0
1
1
0

Q12) B) Light Switch (1, 0)
on off

Q13) D) Q and R $P \rightarrow \text{False}$
Analog more susceptible to noise $\therefore$ any small variation in the s/l can distort the info.
 $Q \rightarrow \text{True}$
Analog s/l's are continuous & can capture an ∞ range of values, resulting in higher resolution
 $R \rightarrow \text{True}$
 $S \rightarrow \text{False}$ Digital signals have limited resolution determined by the bit depth, while analog signals can represent a continuous range of values.

Q14) A)



Q15) A)